

ECO 329 ECONOMIC STATISTICS

Unique Number 36110

MWF 11 am - 12 pm at WAG 101

Fall 2025 Syllabus

1 Key Contact Information

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SI Sessions: M 1 - 2 pm at BEN 1.122 &
F 2 - 3 pm at BEN 1.108

Course Canvas Page <https://utexas.instructure.com/courses/1426197>

The most up-to-date syllabus and most available learning materials are findable on the course Canvas page.

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2 Course Description

2.1 Overview

Welcome to ECO 329, which is an introductory course in probability and statistics and may be your first course in quantitative methods at the upper level of the economics course sequence. The objective of this course is to prepare students for further studies of econometrics by providing students with (1) basic knowledge about interpretation of economic data, (2) an introduction to statistical models, estimation, and inference in economics, and (3) instances involving combinations of theoretical statistics and mathematical background with real-life data analysis skills in economics and business situations. In detail, this course is expected to provide you with:

- Statistics foundation, via exploring methods for describing data;
- Theoretical background, via introducing a variety of statistical methods used to draw conclusions from economic data and exposing students to the conceptual and mathematical foundations of these methods;
- Applicability of theoretical tools, via exploring data analysis skills, e.g., how to build and analyze probability models that can be applied to real-life economic and business situations.

2.2 Prerequisites

- Economics 304K and 304L, with a grade of at least a C- in each;
- Mathematics 408Q, or Mathematics 408K and 408L, or 408C and 408D, or 408C and 408L, or 408N and 408S, or 408K and 408S, or 408C and 408S, or 408L and 408R, or 408R and 408S, or 408L and 408N, with a grade of at least C- in each.

2.3 Activities

You are supposed to be aware of the following course activities - and effectively make use of them - so as to succeed in this course.

- Regularly attend classes.
- Head to office hours for questions.
- Attend weekly SI sessions.
- Finish problem sets in time and submit your answers online before the respective due dates.
- Take in-person exams during your designated exam times.

2.4 Contents

The course materials are divided into the following topics.

- Visual methods of descriptive statistics
- Probability theory
- Discrete random variables
- Continuous random variables
- Sampling theory
- Estimation
- Hypothesis testing

2.5 Schedule

Date	Content	Text References
8/25	Syllabus tour & introduction	1.1 - 1.2
8/27	Descriptive statistics: visual methods	1.2
8/29	Probability and set theory	2.1
9/1	Labor Day, no class	N/A
9/3	Probability and set theory	2.1 - 2.2
9/5	Probability and set theory	2.2
9/8	Conditional probability	2.4
9/10	Conditional probability and independence	2.4 - 2.5
9/12	Counting rules	2.3
9/15	Discrete random variables: basics	3.1, 3.2, 5.1
9/17	Discrete random variables: basics	3.2, 5.1
9/19	Discrete random variables: descriptive statistics	1.3, 1.4, 3.3, 5.2
9/22	Discrete random variables: descriptive statistics	1.4, 3.3, 5.2
9/24	Bernoulli and binomial distributions	3.1, 3.4
9/26	Hypergeometric and Poisson distributions	3.5, 3.6
9/29	Exercise class	N/A
10/1	Midterm 1	N/A
10/3	Continuous random variables: basics	4.1, 4.2, 5.1
10/6	Continuous random variables: descriptive statistics	4.2, 5.2
10/8	Continuous random variables: descriptive statistics	4.2, 5.2, 5.5
10/10	Normal distribution	4.3, 5.5
10/13	Uniform, exponential, t distribution, and chi-squared distribution	4.1, 4.4, 7.3, 7.4
10/15	Sampling and point estimation	5.3, 5.4, 6.1
10/17	Interval estimation: basics and mean via normal (z) distribution	7.1, 7.2
10/20	Interval estimation: mean via normal (z) distribution	7.2
10/22	Interval estimation: mean via t distribution	7.3
10/24	Interval estimation: variance via chi-squared distribution	7.4
10/27	Exercise class	N/A
10/29	Midterm 2	N/A
10/31	Hypothesis testing: basics	8.1
11/3	Hypothesis testing: basics	8.1
11/5	z test for mean	8.2
11/7	z test for proportion	8.4
11/10	t test for mean	8.3
11/12	t test for mean	8.3
11/14	CI method for hypothesis testing and chi-squared test for variance	8.5
11/17	z test for mean difference	9.1
11/19	z test for mean difference	9.1
11/21	z test for proportion difference	9.4
11/24 - 28	Thanksgiving, no classes	N/A
12/1	t test for mean difference	9.2
12/3	pooled z test for mean difference	9.3
12/5	F test for variance ratio	9.5
12/8	Exercise class	N/A
12/11	Final Exam - from registrar's final exam schedule	N/A

Please note that the above course schedule is tentative, aspirational, and describes the desired - but not actual - pace of the course, and I am very likely to fall behind. Thus, please use the schedule mainly as a reference regarding the order in which course materials are to be covered, and I will post announcements for necessary adjustments such as postponements of midterms (if any).

Also, exercise classes may be used to make up for my possibly slow pace, while if I'm sufficiently fast, I may also cover some review questions (e.g., past exam questions) and/or take pop-up questions there.

3 Course Materials and Resources

3.1 Textbook

The textbook for this course is

Jay L. Devore, *Probability and Statistics for Engineering and the Sciences*, 9th Edition, Cengage.

The text references in the course schedule are associated with the 9th edition of the textbook, while you are free to use earlier editions as long as you don't mind mapping the contents in the 9th edition and those in the earlier edition you use. In your exams you are only responsible for contents in lecture notes covered in class - which will also be posted on Canvas - and thus won't be directly responsible for any content in any version of any textbook.

Any way of acquiring the textbook is fine (hard-cover or loose-leaf version or e-book, new or used). For your reference, the materials for this class are available through the Longhorn Textbook Access (LTA) program, an initiative between UT Austin, The University Co-op and textbook publishers to significantly reduce the cost of digital course materials for students. You are automatically opted into the program but can easily opt-out (and back in) via Canvas through the 12th class day (September 10th). To opt out, simply press the "My Textbooks" link in Canvas and then press the "Opt Out" button on the material.

You will be billed for your LTA material after the 5th class day through your "What I Owe" page but still have through the 12th class day to opt out. You must pay for your LTA materials through the What I Owe page by the 20th class day (September 22nd). If you do not pay by the 20th class day, you will lose access to the materials after the 20th class day and your charge will be removed. More information about the LTA program is available at <https://www.universitycoop.com/longhorn-textbook-access>

3.2 Lecture Notes

I will use hand-typed notes (content based on the textbook, while reorganized and typed by myself) for lectures and present them via the projector in class. The notes will be uploaded (and updated if necessary) in Files tab on Canvas and, as is mentioned above, these lecture notes are the only materials that you will be responsible for in your exams.

3.3 Office Hours (Optional)

I will hold a one-hour regular office hours per week in person (time and location findable on the first page of the syllabus). Each of your teaching assistants will also hold a one-hour regular office hour per week (times and locations findable on the first page of the syllabus). As you may have known, office hours are intended to be used for addressing pop-up questions and, contrary to traditional review sessions/exercise classes, there will be no specific materials prepared to be covered in office hours - so be sure to bring your questions there.

3.4 SI Sessions (Optional)

Starting from the second week of the semester, your SI leader will lead two identical one-hour SI sessions in person. In SI sessions, your SI leader will expose you to course contents and extra practice problems, and you will have opportunities to engage in various collaborative learning activities with your classmates.

4 Assessments

4.1 Problem Sets: 15%

The problem set question sheets will be posted in Files tab on Canvas, and you are supposed to submit your answers to each problem set to the corresponding homework assignment on Gradescope, where the problem

sets will be graded. The first problem set will be due on September 12, and from then on, in most - if not all - weeks without midterms there will be problem sets due until the first week in December, in which the last problem set will be due. Each problem set will be due at 5 pm of a Friday. No late submissions will be accepted, and you need to contact me for necessary extensions beforehand - at least one day before the due date of the relevant problem set - unless there is any unexpected sudden medical emergency, in which situation you need to contact me afterwards but in time to discuss possible remediations. Each problem set has a total of 100 points, and the average of your 8 highest problem sets grades will be counted in the calculation of your gross numerical grades.

The answer key to each problem set will be posted in Files tab on Canvas after the grades of the problem set are posted. Meanwhile, because your problem sets will be graded by your teaching assistants, any grading-relevant concerns regarding your problem sets are supposed to be first directed to them.

4.2 Exams: 85% [25% (MT1) + 25% (MT2) + 35% (Final)]

4.2.1 Logistics

The two midterm exams will be 45-min in-class non-cumulative exams, and the final exam will be a 120-min in-person cumulative exam held during the final exam period in December. All exams will have a total of 100 points and will be closed-book and closed-note - necessary probability tables will be included in the exam sheets. You need to bring your own pens (but NOT pencils, for the sake of scanning) to the exam room so that you can clearly write down your own answers on the exam sheets distributed to you. You may bring a simple calculator (but NOT one that is able to store and display text) to use in the exams, but the usage of any other electronic devices is not allowed in the exams.

4.2.2 Grading Criteria, Answer Key, and Possible Regrade Requests

You are supposed to show sufficiently many details of your work to earn points in your exams. You do not have to show every step (e.g., don't hand in your unorganized scratch), but you must show enough work for us (as graders) to follow your logic. For each exam, after grades are posted, you will find rubric items along with your (scanned) graded exams on Gradescope that correspond to the points that you have earned/lost in each exam question. From the rubric you will see that (1) most points are awarded to the intermediate work of yours and (2) calculation errors tend to make you lose relatively few points.

The answer key to each exam will be posted in Files tab on Canvas on the same day when the corresponding grades are posted. For midterms, you may submit your regrade requests on Gradescope in one week after grades are posted - for example, when grades are posted on a Wednesday, you will have until the end of next Tuesday (before 11:59 pm CDT) to request regrading. As for the final exam, because of the tight time constraint of submitting grades to registrar, the grades will be posted no later than 1 pm CDT of a day, and you will have until the end of the same day (before 11:59 pm CDT) to submit your regrade requests on Gradescope. All regrading concerns will be settled no later than the end of the next day, after which all grades of the exam should be considered as final.

For your reference, I have uploaded the exam question sheets and answer keys for the same course I led in this summer in Files tab on Canvas. You may find that some footnotes in the answer keys seem irrelevant to you, and this is because they correspond to the slides used in summer - while I won't use them in this semester - and thus you could ignore such footnotes. My reason for providing these past exams is that you may use the exam questions for practice and learn from the answer keys about how full-point answers are expected to be written and organized.

4.2.3 Policies on Absence and Accommodations

For every one of you:

- There will be NO make-up for any of the exams. If you have any reason for your absence from either (or both) midterm, the weight on the missed midterm(s) will be directly added to that of your final exam in the calculation of your gross numerical grade.

You do NOT need to contact the TAs or me for the weight adjustment regardless of the reason for your absence from the midterm(s). The corresponding weight will be automatically adjusted in the calculation of your gross numerical grade upon the no-submission (or rejection) of any of your midterms - please note that a 0 score of any of your midterms will NOT result in any weight adjustment.

However (and unfortunately), you CANNOT rightfully miss the final exam unless your absence is in accordance to the Class Absence Notification Request or other relevant materials from the office of the dean of the students (<https://deanofstudents.utexas.edu/sos/index.php>).

Please note that it is unlikely for you to gain a decent grade in this course with your unrightful absence from the final exam - such an absence will result in a 0 score of your final exam.

- If you have extended time in your accommodations letter from Disabilities and Access (D&A), upon my receipt of your letter, I will give additional time on each of your exams in Canvas in accordance to your accommodation. You need to email me a PDF of your accommodation letter to have your extended time granted, and I'll reply to initiate the discussion about how the accommodation will be provided.

Please note that students with accommodations are responsible for contacting me directly, providing a PDF of the accommodation letter, letting me know which accommodation they would like to use, and discussing with me how the accommodations will be provided - the notice I receive from D&A that your accommodation letter is available in the portal is NOT sufficient.

4.3 From Raw Grades to Gross Numerical Grade and Letter Grade

- Below is the formula applied to calculate gross numerical grade from raw grades:

$$N = 0.15p + 0.35\hat{f} + \begin{cases} 0.25\hat{m}_1 + 0.25\hat{m}_2 & \text{upon no absence from midterms} \\ 0.25\hat{f} + 0.25\hat{m}_2 & \text{upon absence from midterm 1 only} \\ 0.25\hat{m}_1 + 0.25\hat{f} & \text{upon absence from midterm 2 only} \\ 0.5\hat{f} & \text{upon absence from both midterms} \end{cases},$$

where N is your gross numerical grade,

p is the average of your 8 highest problem set scores,

$\hat{m}_j$ is the adjusted score of your midterm j for each $j \in \{1, 2\}$, and

$\hat{f}$ is the adjusted score of your final exam (each exam will be with 100 points in total).

- In terms of the adjustment of the exam grades, the grades may (or may not) be adjusted in the following manner:

$$\hat{m}_1 = m_1 + a_1, \hat{m}_2 = m_2 + a_2, \hat{f} = f + a_f,$$

where m_1, m_2, f are the raw grades of the corresponding exams, and the adjustment parameters $a_1, a_2, a_f \geq 0$ are associated with the overall performance of the whole class in the corresponding exams. In particular, the effect of these parameters is turning some points of some questions with relatively poor overall performance into bonus points. I will post the overall accuracies of all exam questions along with the grades after each exam, and I will announce the values of a_1, a_2, a_f after ALL grades are posted - and you will eventually find that the adjustments (if any) simply turn the stuff with the poorest overall performance into bonus.

- In terms of your letter grade, if (1) your raw grade is at least ($\geq$) 95 in your final exam, AND (2) you take BOTH midterms, AND (3) you complete ALL problem sets, then you will receive an automatic A for this course.

If the above situation doesn't apply to you, below is the correspondence between the gross numerical grade and the letter grade:

$$\text{Letter Grade} = \begin{cases} A & \text{if } N \geq 95 \\ A- & \text{if } N \in [90, 95) \\ B+ & \text{if } N \in [86, 90) \\ B & \text{if } N \in [83, 86) \\ B- & \text{if } N \in [81, 83) \\ C+ & \text{if } N \in [78, 81) \\ C & \text{if } N \in [74, 78) \\ C- & \text{if } N \in [69, 74) \\ D+ & \text{if } N \in [65, 69) \\ D & \text{if } N \in [62, 65) \\ D- & \text{if } N \in [60, 62) \\ F & \text{if } N < 60 \end{cases} .$$

Please note that, given the correspondence above, you are not competing against each other (your classmates) to earn higher grades, and it is possible for all of you to get decent letter grades if you are with sufficiently high raw grades of problem sets and exams. Thus, you are encouraged to study and work together whenever possible and rightful (e.g., organizing study groups, working together on problem sets) so as to make improvements in a mutually beneficial manner.

Finally, please be aware that your gross numerical grades and letter grades will be posted on Canvas before your letter grades become final and are submitted to the registrar - and you will have time to review and reach out to me for any relevant calculation error before the submission.

5 Other General Academic Guidelines and Resources

- Statement on Academic Integrity

All of your participation in this class is expected to be in accordance with the UT Honor Code:

“The University of Texas at Austin strives to create a dynamic and engaging community of teaching and learning where students feel intellectually challenged; build knowledge and skills; and develop critical thinking, creativity, and intellectual curiosity. As a part of this community, it is important to engage in assignments, exams, and other work for your classes with openness, integrity, and a willingness to make mistakes and learn from them.

The UT Austin honor code champions these principles:

I pledge, as a member of the University of Texas community, to do my work honestly, respectfully, and through the intentional pursuit of learning and scholarship.

The honor code affirmation includes three additional principles that elaborate on the core theme:

- I pledge to be honest about what I create and to acknowledge what I use that belongs to others.
- I pledge to value the process of learning in addition to the outcome, while celebrating and learning from mistakes.
- This code encompasses all of the academic and scholarly endeavors of the university community.

The honor code is more than a set of rules, it reflects the values that are foundational to your academic community. By affirming and embracing the honor code, you are both upholding the integrity of your work and contributing to a campus culture of trust and respect.”

Students who violate University rules on academic misconduct are subject to the student conduct process. A student found responsible for academic misconduct may be assigned both a status sanction and a grade impact for the course. The grade impact could range from a zero on the assignment in question up to a failing grade in the course. A status sanction can range from a written warning, probation, deferred suspension and/or dismissal from the University. To learn more about academic integrity standards, tips for avoiding a potential academic misconduct violation, and the overall conduct process, please visit the Student Conduct and Academic Integrity website at: <http://deanofstudents.utexas.edu/conduct>

- Sharing of Course Materials is Prohibited

No materials used in this class, including, but not limited to, lecture hand-outs, assessments (quizzes, exams, papers, projects, homework assignments), in-class materials, review sheets, and additional problem sets, may be shared online or with anyone outside of the class unless you have my explicit, written permission. Unauthorized sharing of materials may facilitate cheating. The University is aware of the sites used for sharing materials, and any materials found online that are associated with you, or any suspected unauthorized sharing of materials, will be reported to Student Conduct and Academic Integrity in the Office of the Dean of Students. These reports can result in initiation of the student conduct process and include charge(s) for academic misconduct, potentially resulting in sanctions, including a grade impact.

- Use of Email for Official Correspondence to Students

All students should become familiar with the University's official email student notification policy. It is the student's responsibility to keep the University informed of any changes in his or her email address. Students are expected to check email on a frequent and regular basis in order to stay current with University communications, recognizing that certain communications may be time-critical. This includes emails from instructors. It is recommended that email be checked daily.

- Accommodations for Students with Disabilities

The university is committed to creating an accessible and inclusive learning environment consistent with university policy and federal and state law. Please let me know if you experience any barriers to learning so I can work with you to ensure you have equal opportunity to participate fully in this course. If you are a student with a disability, or think you may have a disability, and need accommodations please contact Disability and Access (D&A). Please refer to D&A's website for contact and more information: <http://diversity.utexas.edu/disability/>. If you are already registered with D&A, please deliver your Accommodation Letter to me as early as possible in the semester so we can discuss your approved accommodations and needs in this course.

- Religious Holy Days

By UT Austin policy, you must notify me of your pending absence for a religious holy day as far in advance as possible of the date of observance. If you must miss a class, an examination, a work assignment, or a project in order to observe a religious holy day, you will be given a reasonable compensation or an opportunity to complete the missed work within a reasonable time after the absence.

- Title IX

Title IX is a segment of the Education Amendments of 1972 and makes it clear that violence, harassment, or discrimination based on sex and gender are Civil Rights offenses subject to the same kinds of accountability and the same kinds of support applied to offenses against other protected categories such as race, national origin, sexuality, disability, etc. If you or someone you know has been harassed or assaulted, please don't hesitate to speak to anyone on our teaching team. You can also find useful resources at <https://www.utexas.edu/student-affairs/policies/title-ix>

- Counseling and Mental Health Center

As a student, you may experience a range of challenges that can interfere with learning, such as strained relationships, increased anxiety, substance use, feeling down, difficulty concentrating and/or lack of motivation. These mental health concerns or stressful events may diminish your academic performance and/or reduce your ability to participate in daily activities. Services are available here at UT, and treatment does work. The UT Counseling and Mental Health Center (<http://chmc.utexas.edu/>) provides counseling, psychiatric, consultation, and prevention services that facilitate students' academic and life goals and enhance their personal growth and well-being. The CMHC crisis line is available 24/7/365 at 512-471-2255; our COFA Care Counselor Nathan Langfitt can be reached at the direct line 512-471-0392. A wonderful resource at the CMHC is the MindBody Lab, a self-paced environment designed to help UT students explore various resources for improving their emotional and physical health. The Lab currently features audio and video instruction on a variety of topics, including sleep issues; food and body image; health and well-being; relaxation and meditation. Most of this material is experiential, enabling students to follow along and practice the skills as they are being discussed.

- Emergency Evacuation

From the Office of Campus Safety and Security (<http://www.utexas.edu/safety/>):

- Occupants of buildings on The University of Texas at Austin campus are required to evacuate buildings when a fire alarm is activated. Alarm activation or announcement requires exiting and assembling outside.
- Familiarize yourself with all exit doors of each classroom and building you may occupy. Remember that the nearest exit door may not be the one you used when entering the building.
- Students requiring assistance in evacuation shall inform their instructor in writing during the first week of class.
- In the event of an evacuation, follow the instruction of faculty or class instructors.
- Do not re-enter a building unless given instructions by the following: Austin Fire Department, The University of Texas at Austin Police Department, or Fire Prevention Services office.
- Behavior Concerns Advice Line (BCAL): 512-232-5050
- Link to information regarding emergency evacuation routes and emergency procedures can be found at: utexas.edu/emergency